

ADDITIONAL PROBLEM SHEET 1: SOLUTIONS

1. Practice for solving ODEs

(a) This equation can be separated,

$$\int \frac{1}{y^3} dy = \int \frac{x}{\sqrt{1+x^2}} dx. \quad (1)$$

Integrating both sides we obtain

$$-\frac{1}{2y^2} = \int \frac{x}{\sqrt{1+x^2}} dx + C. \quad (2)$$

To calculate the x -integral we substitute $u = 1 + x^2$, $dx = du/2x$ to obtain

$$\int \frac{x}{\sqrt{1+x^2}} dx = \int \frac{1}{2\sqrt{u}} du = \sqrt{u} = \sqrt{1+x^2} \quad (3)$$

so

$$-\frac{1}{2y^2} = \sqrt{1+x^2} + C. \quad (4)$$

It's best to apply the boundary condition here, so $y(0) = y_0$ implies

$$C = -\left(\frac{1}{2y_0^2} + 1\right) = -\frac{1+2y_0^2}{2y_0^2}. \quad (5)$$

Thus our final solution is the beautiful

$$y = \frac{1}{\sqrt{2\frac{1+2y_0^2}{2y_0^2} - 2\sqrt{(1+x^2)}}}. \quad (6)$$

(b) A common trick when there are logarithms present is to write $y = e^{v(x)}$. Then our equation can be written as

$$x \frac{dv}{dx} = \log(x) - v, \quad (7)$$

which is now a first order ODE where we can use an integrating factor. Writing the ODE as

$$\frac{dv}{dx} + \frac{1}{x}v = \frac{\log(x)}{x}, \quad (8)$$

the integrating factor is $e^{\int (1/x) dx} = x$. Therefore,

$$\frac{d}{dx}[xv] = \log(x) \quad (9)$$

and

$$v = \frac{1}{x}[x \log x - x + C] = \log x - 1 + C/x. \quad (10)$$

Going back to y , recalling $y = e^v$, we therefore have

$$y = xe^{C/x-1} \quad (11)$$

and the boundary condition implies $C = \log(A) + 1$.

(c) We can separate variables in this case,

$$\int \frac{1}{y^2} dy = \int \frac{1}{x^2 + 3x + 2} dx + \frac{C}{\mu}. \quad (12)$$

To perform the x -integral we need to use partial fractions! We factorise the quadratic,

$$x^2 + 3x + 2 = (x + 2)(x + 1). \quad (13)$$

and then separate the fraction,

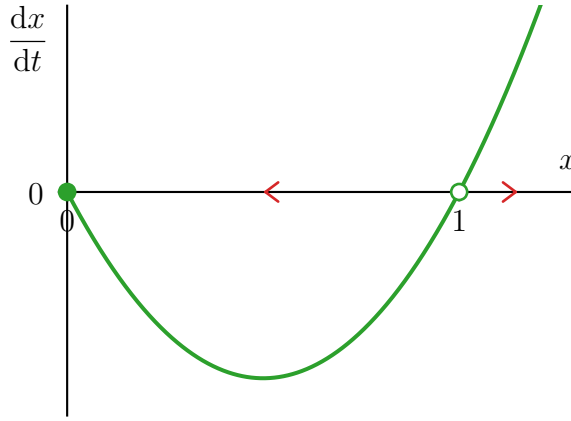
$$\frac{1}{x^2 + 3x + 2} = \frac{A}{x + 2} + \frac{B}{x + 1} \implies A(x + 1) + B(x + 2) = 1, \quad (14)$$

from which we can ascertain that $A = -1$ and $B = 1$. Together then, we have

$$-\frac{1}{y} = -\log(x + 2) + \log(x + 1) + C \implies y = \frac{1}{\log\left(\frac{x+2}{x+1}\right) + C}. \quad (15)$$

2. 1D stability analysis

(a) The equilibria are $x = 0$ and $x = 1$. The plot is



and $x = 0$ is stable, while $x = 1$ is unstable.

Epsilon-style analysis: Let $x = x_0 + \varepsilon x_1$. For this first question we will derive the expansion ourselves. Then the governing equation becomes

$$\varepsilon \frac{dx_1}{dt} = (x_0 + \varepsilon x_1)(x_0 + \varepsilon x_1 - 1). \quad (16)$$

Looking at each order,

$$[1] \quad 0 = x_0(x_0 - 1), \quad (17)$$

$$[\varepsilon] \quad \frac{dx_1}{dt} = x_1(2x_0 - 1). \quad (18)$$

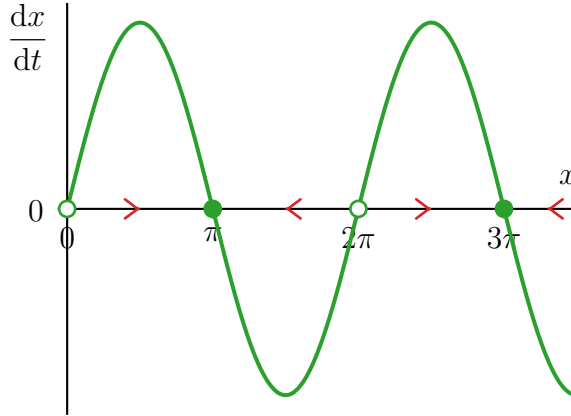
This final $\mathcal{O}(\varepsilon)$ equation is the result from class mentioned on the question sheet. Evaluating this equation at each equilibrium:

$$x_0 = 0 : \quad \frac{dx_1}{dt} = -x_1 \implies x_1 = Ae^{-t} \implies \text{stable}, \quad (19)$$

$$x_0 = 1 : \quad \frac{dx_1}{dt} = x_1 \implies x_1 = Ae^t \implies \text{unstable}, \quad (20)$$

which confirms what we found earlier.

(b) The equilibria are $x = 0, \pi, 2\pi, 3\pi \dots$. The plot is



and we can see that $x = 0$ is unstable, $x = \pi$ is stable, $x = 2\pi$ is unstable, and so on.

Epsilon-style analysis: you should get

$$\frac{dx_1}{dt} = x_1 \cos(x_0). \quad (21)$$

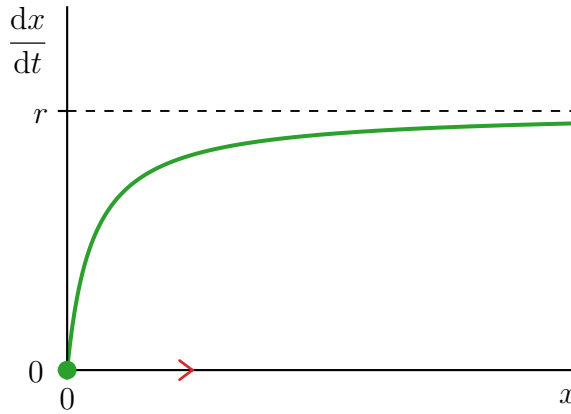
Evaluating this equation at each equilibrium:

$$x_0 = 0 : \quad \frac{dx_1}{dt} = x_1 \quad \implies x_1 = Ae^t \implies \text{unstable}, \quad (22)$$

$$x_0 = \pi : \quad \frac{dx_1}{dt} = -x_1 \quad \implies x_1 = Ae^{-t} \implies \text{stable}, \quad (23)$$

and so on, matching what we had before.

(c) The only equilibrium is $x = 0$. The plot is



and we can see that x is unstable. You can get away without the plot here by noticing that, in fact, $dx/dt > 0$ always!

Epsilon-style analysis: you should get

$$\frac{dx_1}{dt} = \frac{Krx_1}{(K + x_0)^2}. \quad (24)$$

Evaluating this equation at the equilibrium:

$$x_0 = 0 : \quad \frac{dx_1}{dt} = \frac{r}{K}x_1 \quad \implies x_1 = Ae^{rt/K} \implies \text{unstable}, \quad (25)$$

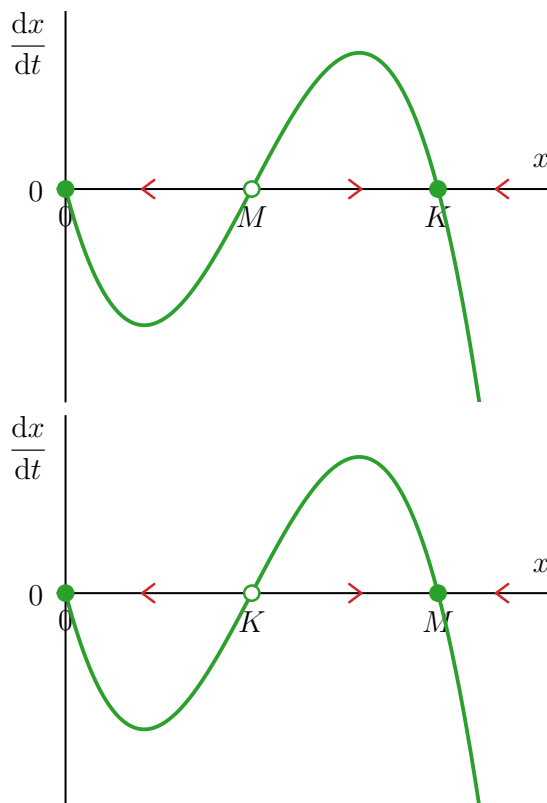
matching what we had before.

3. And fine-Allee... (Exam section B style)

The Allee equation in this form is

$$\frac{dx}{dt} = rx(K - x)(x - M). \quad (26)$$

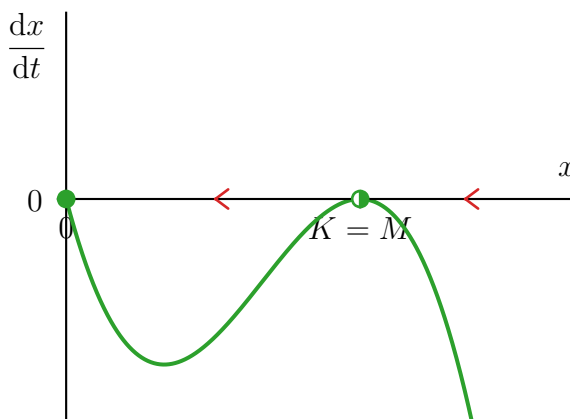
- (a) The equilibria are $x = 0$, $x = K$ and $x = M$.
 (b) The phase plot looks like one of these:



and we can see that, so long as $K \neq M$,

- $x = 0$ is stable,
- $x = \min(K, M)$ is unstable,
- $x = \max(K, M)$ is stable.

There is an interesting question about what happens when $K = M$. Then the plot looks like



and while $x = 0$ remains stable, $x = K = M$ is only stable if we move to the right, while it is unstable if we move to the left. Such a point is called a *saddle point*, and we'll look at these later in the course.

- (c) i. Equation (26) is separable so

$$\int \frac{1}{x(1-x)(x-2)} dx = rt + c, \quad (27)$$

with c a constant of integration. For the integral we use partial fractions

$$\frac{B}{x} + \frac{C}{1-x} + \frac{D}{x-2} = \frac{B(1-x)(x-2) + Cx(x-2) + Dx(1-x)}{x(1-x)(x-2)} \quad (28)$$

$$= \frac{1}{x(1-x)(x-2)}. \quad (29)$$

Letting $x = 0$ gives $B = -1/2$, $x = 1$ gives $C = -1$ and $x = 2$ gives $D = -1/2$. Thus

$$\int \frac{1}{x(1-x)(x-2)} dx = -\int \frac{1}{2x} dx + \int \frac{1}{x-1} dx - \int \frac{1}{2(x-2)} dx \quad (30)$$

$$= -\frac{1}{2} \log(x) + \log(x-1) - \frac{1}{2} \log(x-2), \quad (31)$$

$$= \log \left(\frac{x-1}{\sqrt{(x-2)x}} \right), \quad (32)$$

so

$$\frac{x-1}{\sqrt{(x-2)x}} = Ae^{rt} \implies \frac{(x-1)^2}{(x-2)x} = Ae^{2rt}. \quad (33)$$

We can write this as a quadratic,

$$x^2(1 - Ae^{2rt}) - 2(1 - Ae^{2rt})x + 1 = 0, \quad (34)$$

eventually giving us

$$x(t) = 1 \pm \frac{\sqrt{Ae^{2rt}(Ae^{2rt} - 1)}}{1 - Ae^{2rt}}. \quad (35)$$

- ii. First just note that our stable equilibria, given $1 = K < M = 2$, are $x = 0$ and $x = 2$.

Now, as $t \rightarrow \infty$, $x(t)$ behaves like

$$x(t \rightarrow \infty) \sim 1 \pm \frac{\sqrt{(Ae^{2rt})^2}}{-Ae^{2rt}} \sim 1 \pm (-1) = 0 \text{ and } 2, \quad (36)$$

our stable equilibria.

- iii. We have a singularity when

$$1 - Ae^{2rt} = 0 \implies A = \frac{1}{e^{2rt}}. \quad (37)$$

Given $t > 0$, the possible values for A are therefore $0 < A < 1$.

(d) If $x \approx x_0 + \varepsilon x_1$ then we can expand

$$\frac{dx}{dt} = a_0x + a_1x^2 + a_2x^3, \quad (38)$$

as suggested in the question, to find that to linear order,

$$\frac{dx_1}{dt} = a_0x_1 + 2a_1x_0x_1 + 3a_2x_0^2x_1 = (a_0 + 2a_1x_0 + 3a_2x_0^2)x_1. \quad (39)$$

With $K = 1$ and $M = 2$, we have that $a_0 = -2r$, $a_1 = 3r$, $a_2 = -r$.

Equilibrium 1: $x_0 = 0$

$$\frac{dx_1}{dt} = a_0x_1 \implies x_1 = e^{a_0t} \quad (40)$$

but $a_0 < 0$ here, so the solution is stable.

Equilibrium 2: $x_0 = 1$

$$\frac{dx_1}{dt} = (a_0 + 2a_1 + 3a_2)x_1, \quad (41)$$

and $a_0 + 2a_1 + 3a_2 = r > 0$, so the solution is unstable.

Equilibrium 3: $x_0 = 2$

$$\frac{dx_1}{dt} = (a_0 + 4a_1 + 12a_2)x_1, \quad (42)$$

and $a_0 + 4a_1 + 12a_2 = -2r < 0$, so the solution is stable.

This confirms our conclusions from part (b).

We should emphasise here that this formal ‘epsilon-style’ linear stability analysis is not better or ‘more correct’ than the graphical phase plot method we started with. Rather, it is a more general tool which we are just practising, since we can use it for equations which aren’t exclusively of the form $dx/dt = f(x)$. If in an exam you are asked to do a 1D stability analysis of an equation of the form $dx/dt = f(x)$, which nearly all population models are, graphical is nearly always the best way to go.

4. Time-averaged behaviour of Lotka–Volterra

(a) Substituting in the expression for dx/dt ,

$$\int_{t_0}^{t_0+T} \frac{1}{x} \frac{dx}{dt} dt = \int_{t_0}^{t_0+T} \frac{1}{x} (x - xy) dt = \int_{t_0}^{t_0+T} (1 - y) dt. \quad (43)$$

(b) We have

$$\int_{t_0}^{t_0+T} (1 - y) dt = T - \int_{t_0}^{t_0+T} y dt. \quad (44)$$

As

$$\int_{t_0}^{t_0+T} \frac{1}{x} \frac{dx}{dt} dt = \log(x(t_0 + T)) - \log(x(t_0)) = 0, \quad (45)$$

owing to the periodicity of $x(t)$, the identity derived in part (a) then tells us

$$T - \int_{t_0}^{t_0+T} y dt = 0 \implies T = \int_{t_0}^{t_0+T} y dt, \quad (46)$$

as required.

(c) Using the definition,

$$\bar{x} = \frac{1}{T} \int_{t_0}^{t_0+T} x \, dt \quad \text{and} \quad \bar{y} = \frac{1}{T} \int_{t_0}^{t_0+T} y \, dt. \quad (47)$$

We repeat steps (a) and (b) for dy/dt :

$$\int_{t_0}^{t_0+T} \frac{1}{y} \frac{dy}{dt} dt = \gamma \int_{t_0}^{t_0+T} (x - 1) \, dt \quad (48)$$

and integrating it is clear we have

$$\int_{t_0}^{t_0+T} (x - 1) \, dt = \int_{t_0}^{t_0+T} x \, dt - T. \quad (49)$$

A similar argument to eq. (45) shows

$$T = \int_{t_0}^{t_0+T} x \, dt. \quad (50)$$

Now, inserting eqs. (46) and (50) into eq. (47), we obtain

$$\bar{x} = \bar{y} = 1, \quad (51)$$

as required. Note further that the mean is the same as one of the equilibria of the system.

5. 2D stability analysis

The following solutions find the eigenvalues of the Jacobian explicitly, then use them to determine the stability of the equilibria. You can, of course, also use the trace/determinant method from chapter 3, which may well be quicker. It's up to you!

(a) The system is

$$\frac{dx}{dt} = -x - 2y, \quad (52)$$

$$\frac{dy}{dt} = 4x - 5y. \quad (53)$$

The equilibrium is given by

$$x + 2y = 0, \quad 4x - 5y = 0. \quad (54)$$

There is only one possible solution: $x = y = 0$. Since this equation is already linear we find the stability matrix $\mathbf{J}$ is

$$\mathbf{J} = \begin{pmatrix} -1 & -2 \\ 4 & -5 \end{pmatrix}. \quad (55)$$

$\det(\mathbf{J} - \lambda \mathbf{I}) = 0$ gives $(-1 - \lambda)(-5 - \lambda) + 8 = 0$, for which the solutions are

$$\lambda = -3 \pm 2i. \quad (56)$$

Crucially the real part of λ is negative in both cases, so the equilibrium is stable.

(b) The system is

$$\frac{dx}{dt} = xy \pm y^3, \quad (57)$$

$$\frac{dy}{dt} = x \pm y^3. \quad (58)$$

For the positive case the equilibria are found by solving

$$xy = -y^3, \quad x = -y^3. \quad (59)$$

So clearly $(x, y) = (0, 0)$ is an equilibrium. Also we can obtain $y^2 = y^3$ so $y = 1$, but $x = -1$, so this solution is not valid. There are no other real solutions. The stability matrix is

$$\mathbf{J} = \begin{pmatrix} y_0 & x_0 + 3y_0^2 \\ 1 & 3y_0^2 \end{pmatrix} \quad (60)$$

where x_0 and y_0 are the equilibrium values. In this case we only consider $x_0 = 0$ and $y_0 = 0$ so

$$\mathbf{J} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (61)$$

Pretty dull! This leads to the eigen-equation $\lambda^2 = 0$ so the equilibrium is degenerate.

The second case is a bit more interesting. In addition to the $(0, 0)$ equilibrium there is an equilibrium satisfying $y^2 = y^3$ so $y = 1$ and $x = 1$. The stability matrix is

$$\mathbf{J} = \begin{pmatrix} y_0 & x_0 - 3y_0^2 \\ 1 & -3y_0^2 \end{pmatrix}. \quad (62)$$

As for the previous case the $(0, 0)$ equilibrium is degenerate. The $(1, 1)$ equilibrium gives

$$\mathbf{J} = \begin{pmatrix} 1 & -2 \\ 1 & -3 \end{pmatrix}. \quad (63)$$

So $\det(\mathbf{J} - \lambda \mathbf{I}) = \lambda^2 + 2\lambda - 1$ and

$$\lambda = -1 \pm \sqrt{2}. \quad (64)$$

Both eigenvalues are real and one is positive, so this equilibrium is unstable.

(c) The system is

$$\frac{dx}{dt} = -\sin(x^2 + y^2), \quad (65)$$

$$\frac{dy}{dt} = x - y. \quad (66)$$

The equilibria are when

$$x^2 + y^2 = n\pi, \quad x = y \quad (67)$$

so the valid solutions are $x = y = \sqrt{n\pi/2}$. The stability matrix is

$$\mathbf{J} = \begin{pmatrix} -2 \cos(x_0^2 + y_0^2)x_0 & -2 \cos(x_0^2 + y_0^2)y_0 \\ 1 & -1 \end{pmatrix}. \quad (68)$$

We now evaluate this at the equilibria for which $x^2 + y^2 = n\pi$, for which $\cos(n\pi) = \pm 1$ depending on whether n is odd (-1) or even ($+1$).

In the n even case,

$$\mathbf{J} = \begin{pmatrix} -\sqrt{2n\pi} & -\sqrt{2n\pi} \\ 1 & -1 \end{pmatrix} \quad (69)$$

and $\det(\mathbf{J} - \lambda \mathbf{I}) = 0$ gives

$$\lambda^2 + (1 + \sqrt{2n\pi})\lambda + 2\sqrt{2n\pi} = 0. \quad (70)$$

Therefore (horrible, isn't it?),

$$\lambda = \frac{1}{2} \left[-(1 + \sqrt{2n\pi}) \pm \sqrt{(1 + \sqrt{2n\pi})^2 - 8\sqrt{2n\pi}} \right]. \quad (71)$$

As $-(1 + \sqrt{2n\pi}) < 0$ and the root is either $< (1 + \sqrt{2n\pi})$ in magnitude or imaginary (for $n > 0$), the eigenvalues always have a negative real part. Thus the equilibria are asymptotically stable for all even $n > 0$.

For $n = 0$, the eigenvalues are $\lambda = 0$ and -1 , leaving us in the degenerate case.

For the n odd case, $\det(\mathbf{J} - \lambda \mathbf{I}) = 0$ gives

$$\lambda^2 + (1 - \sqrt{2n\pi})\lambda - 2\sqrt{2n\pi} = 0. \quad (72)$$

Therefore,

$$\lambda = \frac{1}{2} \left[-(1 - \sqrt{2n\pi}) \pm \sqrt{(1 - \sqrt{2n\pi})^2 + 8\sqrt{2n\pi}} \right]. \quad (73)$$

There will therefore always be a positive eigenvalue (noting that $n > 0$ for these odd modes) and the equilibria are asymptotically unstable.

6. Proof of periodicity of the solutions to Lotka–Volterra

Our function C is

$$C(x_0, y_0) = \log(y_0) - y_0 + \gamma(\log(x_0) - x_0). \quad (74)$$

For the sake of notational simplicity we set $x = x_0$ and $y = y_0$.

(a) We take partial derivatives to first and second order:

$$\frac{\partial C}{\partial x} = \gamma \left(\frac{1}{x} - 1 \right), \quad \frac{\partial C}{\partial y} = \frac{1}{y} - 1, \quad (75)$$

$$\frac{\partial^2 C}{\partial x^2} = -\gamma \frac{1}{x^2}, \quad \frac{\partial^2 C}{\partial y^2} = -\frac{1}{y^2}, \quad \frac{\partial^2 C}{\partial x \partial y} = 0. \quad (76)$$

The potential critical points in the surface are at $\partial C / \partial x = \partial C / \partial y = 0$ so there is only one possibility, $x = y = 1$. The sign of the function

$$D = \frac{\partial^2 C}{\partial y^2} \frac{\partial^2 C}{\partial x^2} - \frac{\partial^2 C}{\partial x \partial y} \quad (77)$$

tells us what kind of critical point we have when x and y take on their critical values:

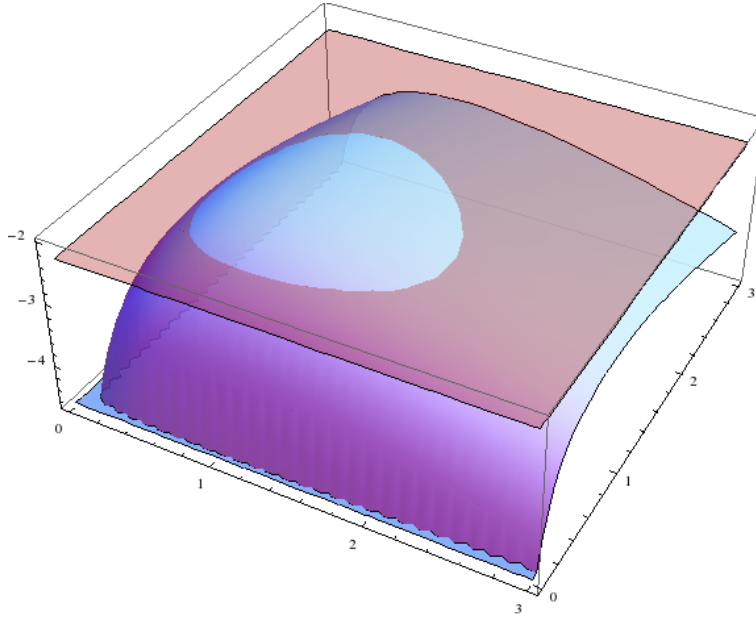


Figure 1: The function f given by (74). Solutions are curves on this surface of constant C , i.e. the intersection of the plane $C = \text{const.}$ and the surface $f(x, y)$. Here we see clearly their intersection will always be a closed curve, hence periodic solutions.

- $D > 0$ and $\partial^2 C / \partial x^2 < 0$: (x_0, y_0) is a maximum,
- $D > 0$ and $\partial^2 C / \partial x^2 > 0$: (x_0, y_0) is a minimum,
- $D < 0$: (x_0, y_0) is a saddle point,
- $D = 0$: indeterminate because $\partial C / \partial x$ and $\partial C / \partial y$ are not linearly independent.

In our case, $D > 0$ and $\partial^2 C / \partial x^2 < 0$ so we have a maximum. As it is the only critical point is a global maximum.

- (b) We can think of C as a surface function on the domain $x_0 \in [0, \infty]$ and $y_0 \in [0, \infty]$ (it is shown in fig. 1 for $\gamma = 1$). The solutions $x(t), y(t)$ are curves $\gamma(t)$ on this surface for which $z = C(x_0, y_0)$ is constant: the ‘height’ must be constant. For a given constant C_0 , a curve (parametrised for a sufficient time period) will be the intersection of the plane $z = C_0$ and the surface of $z = C(x_0, y_0)$. Since the surface has a single global maximum, it looks like a hill, and the intersection of the plane and surface will be a closed curve. A closed curve implies periodicity, but we first must determine that the curve is transversed with a consistent orientation (in a consistent direction around the curve).

Again dropping back to losing the subscript 0s, we consider the curve as a function of x by solving eq. (74) for y given a value of C and x . This gives us a parametrisation for our closed curve $(x, y(x))$. We also remind ourselves that the equations take the form

$$\frac{dx}{dt} = x(1 - y), \quad (78a)$$

$$\frac{dy}{dt} = \gamma y(x - 1). \quad (78b)$$

There are two points at which $dx/dt = 0$, for which we have $y = 1$. If dx/dt changes from increasing to decreasing about $dx/dt = 0$ then y must be increasing

from $y < 1$ to $y > 1$, and if dx/dt changes from decreasing to increasing about dx/dt , then y must be decreasing from $y > 1$ to $y < 1$. This gives the curve the same orientation at the two extremes of the curve in the x direction.

A similar argument for points at which $dy/dt = 0$ (and hence $x = 1$) shows the curve is similarly well oriented at both extremes in y .

In between these points we can consider four quadrants, each with varying signs of dy/dt and dx/dt . For example, let's consider the quadrant $dy/dt > 0$ and $dx/dt > 0$. If we consider travelling in an anti-clockwise direction we enter this section at a point where $dy/dt = 0$ and for which the derivative dy/dt has changed from decreasing to increasing. In this case x must have increased above 1 and must stay above 1 until it reaches the next point where $dy/dt = 0$. Thus the orientation remains consistent on the quadrant $dy/dt > 0$ and $dx/dt > 0$. A similar argument can be applied on all four quadrants to show that the curve $(x, y(x))$ is consistently oriented, and hence the solutions must be periodic. It is easy to see the same argument would apply if the orientation were reversed.

7. Variations on Lotka–Volterra (Exam 2021 & 2022)

The matchings (with some example reasonings) are:

- (G1)–(P4)–(M3) Logistic. Between (M3) and (M4), (M3) is the only one that allows y to be more than 1, given the initial conditions. For $y > 1$, in (P4), $dy/dt > 0$ in places. This is only possible in (M3) because in (M4), $dy/dt < 0$ for $y > 1$.
- (G2)–(P3)–(M1) Traditional Lotka–Volterra; periodic.
- (G3)–(P2)–(M4) Logistic. Process of elimination.
- (G4)–(P1)–(M2) dy/dt is always negative. Exponential growth of x once y has decayed exponentially. No equilibrium.